AI-BASED FAKE PROFILE DETECTION IN SOCIAL NETWORKS

DETAILED MAJOR PROJECT REPORT

ABSTRACT

Social networking platforms have become essential digital communication environments, but their large scale also creates opportunities for fake, automated, impersonation, spam, and malicious accounts. Fake profiles may be used to conduct scams, distribute unwanted content, manipulate engagement, impersonate individuals, or establish deceptive relationships. Manual identification is difficult because sophisticated fake profiles can imitate legitimate users.

This project proposes an Artificial Intelligence based fake profile detection system that combines profile attributes and behavioral intelligence. The system collects relevant account characteristics, cleans and transforms the data, engineers meaningful behavioral features, and trains supervised machine-learning models to classify accounts as genuine or suspicious. Candidate algorithms include Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. The final model produces a probability and an interpretable risk level.

The system is designed as a decision-support tool. Instead of relying only on a binary prediction, it can present the probability, major contributing features, and recommended review priority. This improves transparency and allows human analysts to investigate high-risk accounts. The architecture can be implemented using Python and Scikit-learn for machine learning, FastAPI for model serving, React for the dashboard, and PostgreSQL or MongoDB for persistence.

Keywords: Artificial Intelligence, Machine Learning, Fake Profile Detection, Behavioral Intelligence, Social Networks, Classification, Risk Scoring, Explainable AI.

CHAPTER 1 – INTRODUCTION

1.1 Background

Social networks allow users to create profiles, communicate, share media, follow other accounts, and participate in online communities. The openness of these systems is valuable but also creates an attack surface. Fake accounts can be created quickly and at low cost, making manual verification difficult.

1.2 Problem Statement

The problem is to develop a system capable of identifying profiles that exhibit characteristics associated with fake or suspicious accounts. The system should use multiple signals rather than a single rule and should provide an understandable risk score.

1.3 Need for the Project

Rule-based detection can fail when attackers change their behavior. A machine-learning system can learn relationships between multiple features and adapt as new labeled data become available. Behavioral intelligence is particularly useful because it examines how an account operates rather than depending only on information entered in the profile.

1.4 Objectives

• Develop a machine-learning based fake-profile classifier.

• Perform data cleaning and exploratory analysis.

• Engineer profile and behavioral features.

• Compare multiple classification algorithms.

• Evaluate models using precision, recall, F1-score and ROC-AUC.

• Provide an interpretable risk score.

• Design an API and dashboard architecture for practical deployment.

• Support human review rather than making irreversible decisions automatically.

1.5 Scope

The project focuses on account-level classification using structured profile and activity information. It can later be expanded to graph analysis, text analysis, image similarity, streaming detection, and continual learning.

CHAPTER 2 – EXISTING AND PROPOSED SYSTEM

2.1 Existing System

Existing methods include manual reports, fixed rules, CAPTCHA, email/phone verification, IP/device signals, and moderation workflows. These methods remain useful but can be bypassed or may generate false positives when used alone.

2.2 Limitations

• Difficulty handling rapidly changing attacker behavior.

• Dependence on manually designed rules.

• Limited ability to combine many weak signals.

• High human-review workload.

• Difficulty explaining complex suspicious behavior at scale.

2.3 Proposed System

The proposed system uses a complete ML pipeline: data collection, preprocessing, feature engineering, model training, evaluation, inference, risk scoring, explanation, and review.

CHAPTER 3 – LITERATURE REVIEW

Research on fake-account detection has explored profile-based classification, behavioral analysis, social-network graphs, anomaly detection, and deep learning. Profile-based systems use attributes such as account age, follower count, following count, profile completeness, and activity volume. Behavioral systems examine posting frequency, interaction patterns, burst activity, and temporal regularity. Graph-based approaches analyze relationships among users and can detect coordinated communities.

A practical gap exists between highly specialized research prototypes and deployable academic systems. This project addresses that gap by combining feature engineering, ensemble classification, explainability, API deployment, and a review-oriented dashboard.

CHAPTER 4 – DATASET AND DATA PREPROCESSING

The selected dataset should contain a target label indicating fake/genuine status and a useful set of profile or behavioral attributes. The exact dataset and license must be documented before publication.

Preprocessing includes duplicate removal, missing-value treatment, type conversion, outlier inspection, categorical encoding, numerical scaling where required, and class-balance analysis.

CHAPTER 5 – FEATURE ENGINEERING

Potential features include:

• Account age in days.

• Followers count.

• Following count.

• Follower/following ratio.

• Number of posts.

• Posts per day.

• Profile completeness.

• Verification indicator.

• Username length.

• Biography length.

• Number of profile changes.

• Average interaction rate.

• Burst activity score.

Derived features can reveal suspicious combinations. For example, a very new account with unusually high activity or an extreme following-to-follower ratio may warrant additional review. Such signals should not be treated as universal proof of malicious behavior.

CHAPTER 6 – MACHINE LEARNING METHODOLOGY

Logistic Regression is used as a baseline because it is simple and interpretable. Decision Trees provide nonlinear decision boundaries and readable rules. Random Forest combines multiple trees and can be robust against noise. Gradient Boosting and XGBoost can model more complex relationships.

The dataset should be divided into training, validation, and testing sets using stratification where appropriate. Hyperparameters should be tuned using cross-validation without leaking test information.

CHAPTER 7 – MODEL EVALUATION

Important metrics are Accuracy, Precision, Recall, F1-score, ROC-AUC and the confusion matrix. Accuracy alone can be misleading on imbalanced datasets.

Precision measures the proportion of predicted fake accounts that are actually fake. Recall measures the proportion of actual fake accounts detected. F1-score balances precision and recall. ROC-AUC measures ranking performance across thresholds.

Final numerical results must be obtained from the actual selected dataset and experiment. No performance value should be claimed without measurement.

CHAPTER 8 – SYSTEM ARCHITECTURE

The architecture contains:

1. Dashboard

2. REST API

3. Validation layer

4. Feature-engineering service

5. ML inference service

6. Risk-scoring service

7. Explanation service

8. Database

9. Monitoring and logging

Data flow:

User → Dashboard → API → Validation → Feature Engineering → ML Model → Risk Scoring → Explanation → Dashboard/Database.

CHAPTER 9 – DATABASE DESIGN

Suggested entities are Profile, Prediction, Explanation, Review, and User. A profile can have multiple prediction records. Each prediction can contain multiple feature-contribution records. Review records can store analyst decisions and comments.

Privacy should be considered throughout the design. Direct identifiers should be minimized, access should be restricted, and retention rules should be defined.

CHAPTER 10 – IMPLEMENTATION

The ML pipeline can be developed in Python using Pandas, NumPy and Scikit-learn. A trained model can be serialized using Joblib. FastAPI can expose a /predict endpoint. React can provide the analyst dashboard. PostgreSQL or MongoDB can store prediction and review information.

CHAPTER 11 – TESTING

Testing should cover preprocessing, feature calculations, model loading, API validation, prediction responses, database operations, authentication, error handling, and dashboard usability. Security testing should include malformed inputs and unauthorized requests.

CHAPTER 12 – RESULTS AND DISCUSSION

The final report should include a model comparison table with Accuracy, Precision, Recall, F1-score, ROC-AUC, training time, and inference time. The best model should be selected using both statistical performance and operational requirements.

CHAPTER 13 – IMPACT

Research impact includes behavioral intelligence and explainable AI. Industry impact includes moderation and fraud-risk prioritization. Societal impact includes reducing exposure to scams and impersonation. Relevant SDGs include SDG 9, SDG 11 and SDG 16.

CHAPTER 14 – FUTURE SCOPE

Future versions can incorporate Graph Neural Networks, transformer-based text analysis, profile-image similarity, real-time stream processing, continual learning, model-drift monitoring, adversarial robustness, and human-in-the-loop retraining.

CONCLUSION

The proposed system demonstrates how AI can combine profile and behavioral signals to identify suspicious social-network accounts. Its modular architecture supports future improvements while maintaining interpretability and human oversight.